

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Software Quality Assurance



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can talk about quality assurance.
- I can explain what quality assurance is.
- I can describe key features about quality assurance.

Ask your partner:

- How do you define an excellent program?
- What do programmers do to provide the best software possible?





Let's Explore!

Discussing Software Quality Assurance (SQA)



Instructions: Read the text below.

- Javier:** Hi, Pablo. I didn't come to the lecture yesterday. I usually don't miss any of them. What did they talk about?
- Pablo:** Mr. Saenz, the CEO, talked about Software Quality Assurance, or SQA.
- Javier:** In my department, we occasionally hear about SQA, but we are never sure of what it is. Did you get a clear picture of what SQA is?
- Pablo:** Well, I always take notes during lectures. Sure, I can tell you. SQA is the process that monitors software creation in order to avoid errors or bugs in the final product. SQA usually covers all the parts of the software production development. The supervision methods used in SQA correspond to international quality standards. On the other hand, these methods sometimes adopt verified software construction models.
- Javier:** That is very interesting. I thought SQA wasn't that important, but now I see its relevance in bug detection. What else do you remember?
- Pablo:** Certain amount of testing and debugging is usually performed while the code is being written. However, a more extensive and systematic testing is always performed when a preliminary version of the program is finished. The priority of testing varies depending on many factors. For instance, the complexity of the software, competitors taking advantage of problems with the program, the costs of new testing tools, etc.
- Javier:** It sounds like a complex process, with many steps and factors to consider.
- Pablo:** It is a complex process. For that reason, one of the ultimate goals of SQA experts is developing artificially intelligent automatic testing software, with the ability to analyze an application from its development stage and carry out systematic tests until its final version.
- Javier:** Thanks Pablo, your summary was very useful.

**Let's Practice!**

SQA functions

Instructions: Underline all the statements with adverbs of frequency in them. Read the text aloud. Check your pronunciation with your instructor.

Software programs always have to be tested for bugs and errors. Quality assurance is very important for manufacturing companies. It follows governmental standards and is only for public entities. These bugs and errors occasionally affect computers and services in a great way. Software quality assurance usually helps avoid the errors and bugs.

**Let's Practice!**

SQA's importance

Instructions: Proofread the sentences. Correct the mistakes on the lines provided.

1. Software quality assurance reduces software meets and complies with standard quality specifications.

2. SQA rejects the developed software to meet the desired quality.

3. SQA processes makes the quality in each step of the development until the software is complete.



Extension Activity

Quality standards in Colombia

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Investigate about SQA in the ISO 9001 in Colombia.
- Take notes of the most relevant information.
- Make a presentation about the key features of SQA in the ISO 9001.
- Share the presentation in class to your teacher and classmates.

Self-Assessment

Yes No

- I investigate about SQA and other quality standards
- I write about quality standards
- I make a presentation about quality standards
- I talk about quality standards

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- CEO** (abbreviation): chief executive officer, manager
e.g. Mr. Saenz, the CEO, talked about Software Quality Assurance, or SQA.
- bug** (noun): error or mistake, especially in software
e.g. I thought SQA wasn't that important, but now I see its relevance in bug detection.
- preliminary** (adjective): before the final part or version, preparatory
e.g. However, a more extensive and systematic testing is always performed when a preliminary version of the program is finished.
- to get a clear picture of** (expression): to understand a topic clearly
e.g. Did you get a clear picture of what SQA is?
- monitor** (verb): to control or supervise
e.g. SQA is the process that monitors software creation in order to avoid errors or bugs in the final product.
- ultimate** (adjective): maximum, last
e.g. One of the ultimate goals of SQA experts is developing artificially intelligent automatic testing software.



Learning Points

Simple Present and Simple Past Tense (review)

We use the **simple present** when talking about actions in the present or facts. On the other hand, we use the **simple past** when talking about complete actions in the past.

We use the **simple present** tense to refer to an action that is repeated or usual. This action can be a habit, a hobby, or a daily event. We can also use this tense to talk about something we often forget or normally don't do.

Examples:

*I **read** about video game programming every day.*

*He **does not write** practical code.*

***Do** you **know** anything about software debugging?*

In addition, we also use the **present simple** to talk about facts or generalizations.

Example: *Computers **perform** automatic tasks all the time.*

In contrast, we use the **simple past** to refer to an action that started and finished at a specific time in the past, even if the specific time is not mentioned. This tense is also used to talk about habits in the past.

Examples:

*We **studied** Java last semester.*

*They **didn't include** all the software features.*

***Did** she correct the subroutine errors?*

Additionally, we can also use the **past simple** to talk about past facts or generalizations that are no longer true.

Example: *Modems **connected** to the internet using copper landlines.*

Examples

- *I always make notes during lectures.*
- *I usually don't miss any of them.*
- *What else do you remember?*
- *Mr. Saenz, the CEO, talked about Software Quality Assurance, or SQA.*
- *I didn't come to the lecture yesterday.*
- *What did they talk about?*



Learning Points

Adverbs of frequency (review)

We use **adverbs of frequency** to describe how often or frequently we do an activity.

Adverbs of frequency are placed before the main verb (except with the verb **to be**):

Subject + adverb of frequency + principal verb + complement

Example: *They **usually** correct errors before running the program.*

When we the verb **to be** is the main verb of a sentence, the position of the **adverb of frequency** changes:

Subject + verb to be + adverb of frequency + complement

Example: *She **is never** satisfied with the final version of the code.*

Take a look at the chart of **adverbs of frequency**:

Adverbs of Frequency	
100%	Always
90%	Usually
80%	Generally
70%	Often
50%	Sometimes
30%	Ocasionally
10%	Hardly ever
5%	Rarely
0%	Never

Examples

- *However, a more extensive and systematic testing is always performed when a preliminary version of the program is finished.*
- *Certain amount of testing and debugging is usually performed while the code is being written.*
- *These methods sometimes adopt verified software construction models.*
- *In my department, we occasionally hear about SQA, but we are never sure of what it is.*